

Behavioral Biometrics- Seminar

How people type, move, and hold a device — and how that becomes an identity signal

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The gap behavioral biometrics fills

Point-in-time authentication assumes one successful login is enough to trust a session for its full duration. Credentials, OTPs, and even device binding all verify a moment, not a person.

Credentials are portable

Phished, purchased, or reused passwords transfer perfectly to an attacker. The login looks legitimate because it is.

Sessions are hijackable

Token theft, malware, and remote-access tooling put an adversary inside an already-trusted session.

Users can be coerced

In scam and social-engineering fraud the genuine user performs the transaction under an attacker's direction.

Physiological vs. behavioral

Physiological biometrics measure what a body is. Behavioral biometrics measure how a person acts — passively, continuously, and without an enrolment ceremony.

	Physiological	Behavioral
Measures	Fingerprint, face, iris, vein	Typing rhythm, cursor path, swipe, gait
Capture	Explicit, at a checkpoint	Passive, throughout the session
Friction	User action required	Invisible to the user
Revocability	Immutable — cannot be reissued	Template can be re-enrolled and drifts over time
Best role	Strong point-in-time proof	Continuous risk signal and fraud detection

Signal modalities

Keystroke dynamics

Dwell and flight times, digraph/trigraph latencies, error and backspace habits, shift-key handedness.

Mouse & pointer dynamics

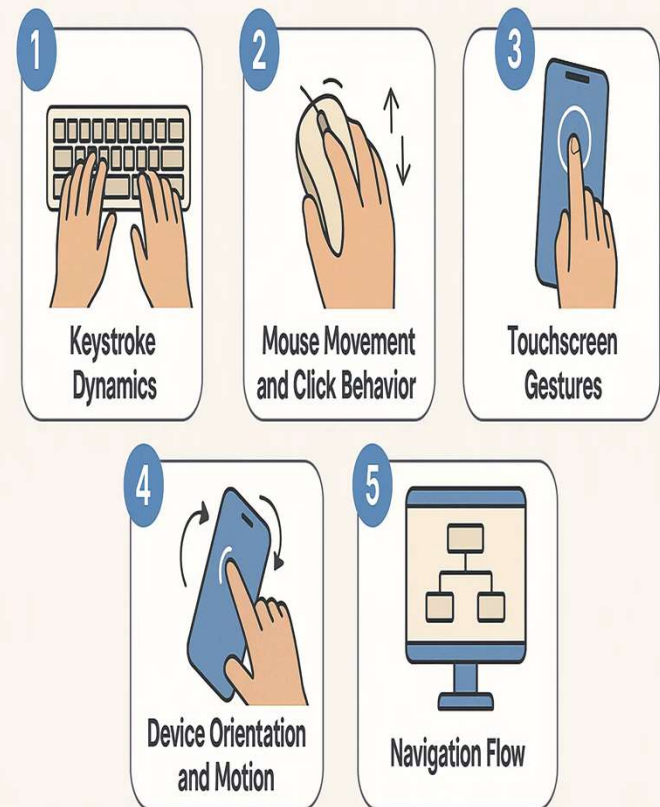
Trajectory curvature, velocity and acceleration profiles, pauses, click pressure and overshoot.

Touch & gesture

Swipe length, pressure, contact area, scroll velocity, pinch-zoom geometry, tap cadence.

Device handling & gait

Accelerometer and gyroscope tilt, grip stability, stride signature from walking motion.



From raw events to features

Capture, feature extraction, and classification are the three modules common to every behavioral system. Most of the engineering effort sits in the middle stage.

01

Capture

- Event stream from SDK or JS agent
- Timestamps at millisecond resolution
- Sampling without keylogging content

02

Extract

- Statistical: mean, variance, percentiles
- Temporal: n-gram latencies, sequences
- Spatial: curvature, jerk, angular change

03

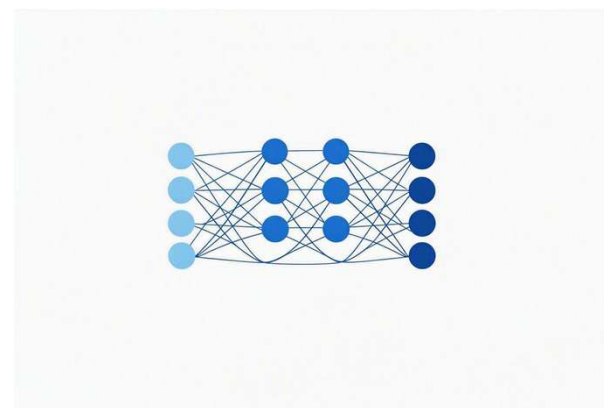
Represent

- Fixed-length feature vectors per window
- Normalization for device and context
- Embeddings from sequence encoders

Design constraint: features must be discriminative across users, stable within a user, and cheap enough to compute at session speed.

Modeling approaches

Distance & statistical	Manhattan/Mahalanobis scoring against an enrolled template. Cheap, interpretable, weak against skilled mimicry.
Classical ML	SVM, random forest, k-NN on engineered features. Strong baselines; needs negative samples or synthetic impostors.
One-class / anomaly	Autoencoders, isolation forests, one-class SVM. Trained only on the genuine user — the realistic production setting.
Deep sequence models	RNN, LSTM, temporal CNN, and transformers over raw event streams. Best accuracy, highest data and latency cost.
Multimodal fusion	Score- or feature-level fusion of keystroke, mouse, touch, and motion. Consistently outperforms any single modality.



Model drift is the operational problem: typing behavior changes with device, injury, fatigue, and time. Templates need scheduled re-enrolment.

Continuous authentication architecture

Risk is scored per action, not once per session. Behavioral signals feed a scoring engine that raises friction only when confidence drops.



Zero Trust fit

Continuous verification replaces the implicit trust granted after a single successful login.

Standards status

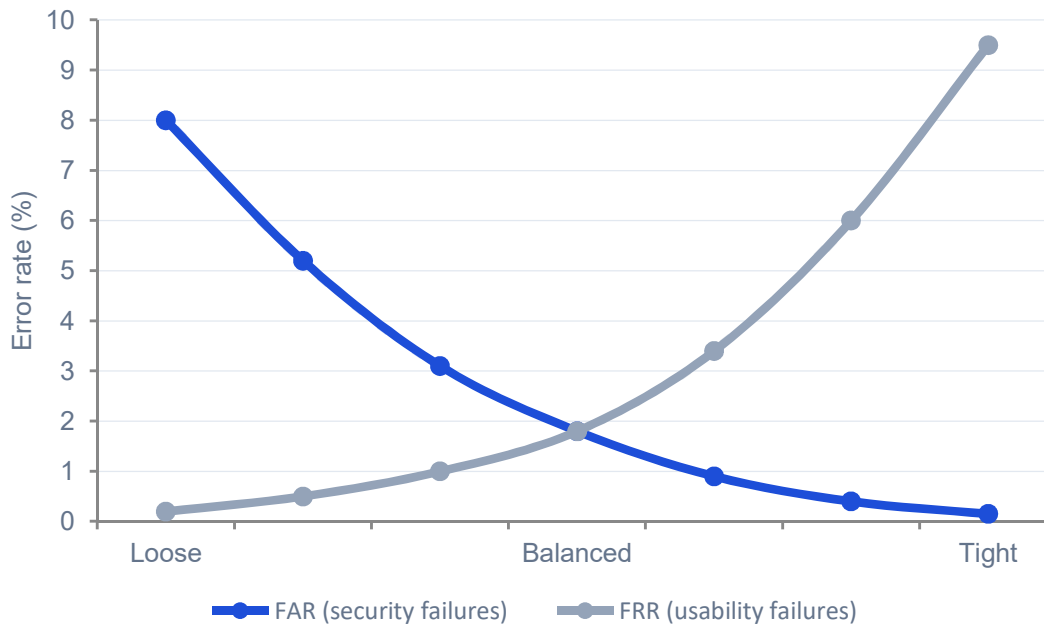
NIST SP 800-63B does not yet codify continuous authentication; risk-based reauth sits alongside AAL, not in place of it.

Latency budget

Scoring must return inside the request path — typically tens of milliseconds — or it will be bypassed.

Accuracy: FAR, FRR, and EER

Every deployment is a threshold decision. Tightening the threshold cuts false acceptances and raises false rejections; the crossover point is the Equal Error Rate.



FAR False Acceptance Rate

Share of impostor attempts wrongly accepted. The security-failure metric.

FRR False Rejection Rate

Share of genuine users wrongly rejected. The usability-failure metric.

EER Equal Error Rate

The point where FAR equals FRR — the standard single-number benchmark for comparing systems.

Threshold choice is business policy: high-assurance banking targets a FAR near one in a million; consumer apps accept roughly one in fifty thousand.

Fraud and security use cases

Account takeover

Login attack rates rose sharply through 2025; behavior separates the credential holder from the credential owner.

Bot and automation detection

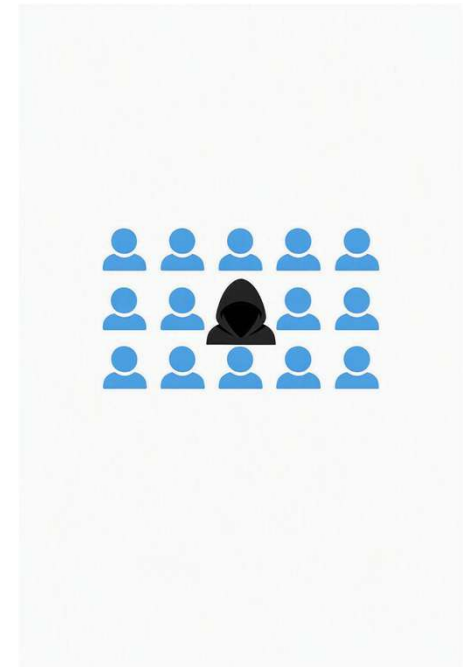
Scripted input has no human variance — near-constant latencies and linear cursor paths are trivially separable.

Remote access and scam coercion

Session hesitation, unusual navigation, and RAT-driven input reveal a user acting under instruction.

New-account and application fraud

Copy-paste of personal data, field re-entry, and unfamiliarity with one's own details flag synthetic identities.



Which signals expose which attack

Attack pattern	Dominant tell	Primary modality
Credential stuffing at scale	Identical timing across sessions and accounts	Keystroke + mouse
Human-operated takeover	Template mismatch on familiar fields	Keystroke + touch
Remote access tooling	Latency artifacts and non-native cursor motion	Mouse dynamics
Coerced or scam-driven transfer	Long hesitation, off-pattern navigation, re-reading	Session behavior
Synthetic identity onboarding	Pasted PII, corrections in own-name fields	Form interaction

No single signal is decisive. Production systems fuse behavior with device, network, and consortium risk intelligence before acting.

Deployment realities

Cold start

A template needs enough genuine sessions before it can score. Plan a shadow-mode period per user, not per deployment.

Context normalization

Desktop, mobile, and tablet produce incompatible feature distributions. Score per device class or normalize explicitly.

False-positive economics

Every point of FRR converts to abandoned sessions and call-centre load. Measure it as a business cost, not a model metric.

Drift and re-enrolment

Behavior shifts with device, posture, injury, and time. Recent work shows accuracy degrading measurably without adaptive templates.

Edge vs. cloud scoring

On-device inference limits data egress and helps privacy posture; central scoring gives cross-account intelligence.

Adversarial robustness

Synthetic input generators and replay of captured event streams are the emerging attack surface — test against them.

Privacy and compliance

GDPR Article 9

Behavioral data used to uniquely identify a person is special-category. Requires an explicit legal basis and, in practice, a DPIA.

Illinois BIPA

Carries a private right of action — the highest litigation exposure of any biometric regime. Written consent and a published retention schedule.

CCPA / CPRA

Biometric identifiers are sensitive personal information, with limitation and deletion rights attached.

EU AI Act

Brings additional obligations for high-risk biometric systems as provisions phase in.

Educational summary — confirm obligations with privacy counsel for your operating jurisdictions.

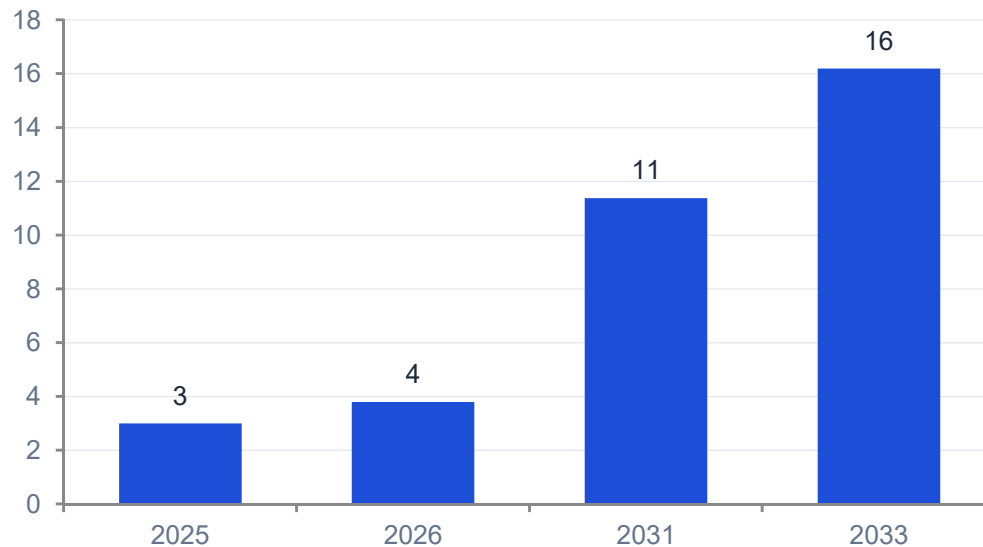


Engineering principles

- Store templates, never raw content
- Capture timing, not keystroke text
- Minimize, retain briefly, honor deletion

Market outlook

Analyst estimates differ widely on absolute size — segment definitions vary — but converge on a high-growth trajectory in the 19-27% CAGR range.



Grand View Research and Mordor Intelligence estimates; 2031 and 2033 figures come from different forecast models.

Software leads

Software accounted for over 66% of spend in 2025; services remain the smaller share.

BFSI dominates demand

Banking and financial services was the largest end-use segment at roughly 24% of revenue.

North America ahead, APAC fastest

North America held over 32% of 2025 revenue; Asia Pacific is the fastest-growing region.

Named vendors

BioCatch, BehavioSec, NuData Security, Zighra, SecuredTouch, and LexisNexis Risk Solutions among others.

What to take away

01 It is a risk signal, not a gate

Behavior belongs in a score that drives step-up, not in a binary allow/deny decision on its own.

02 Fusion beats single modality

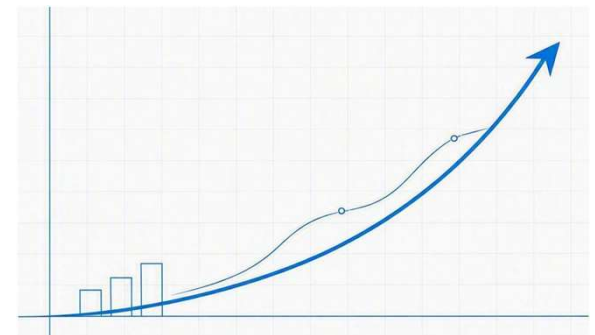
Combine keystroke, pointer, touch, and motion — and combine those with device and network context.

03 Tune to the business, then measure

Set the threshold from fraud appetite and friction tolerance; track EER, but manage FRR as a revenue metric.

04 Build privacy in at capture

Timing over content, templates over raw streams, short retention, and a documented legal basis.



Next step

Run a shadow-mode pilot on one high-value flow — login or payment — and baseline FAR, FRR, and step-up volume before enforcing anything.

Thank you

